# 03 21826 Autumn 2025
# LH The Life and Death of Stars

## Lecture 14

(JCD 10)

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# Previous lecture

- Hydrogen burning happens through pp-chains or CNO cycle
- Helium burning happens through the triple alpha process
- More advanced burning follows burning C, Ne, and O, in that order
- Equations of stellar structure and evolution to describe the variables $m$ or $r, P, L, T, X$
- Additional relations, boundary conditions and initial conditions then used to numerically solve them

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# Outline

Star Formation

Jeans instability

Early stellar evolution

Approach to the main sequence

Summary

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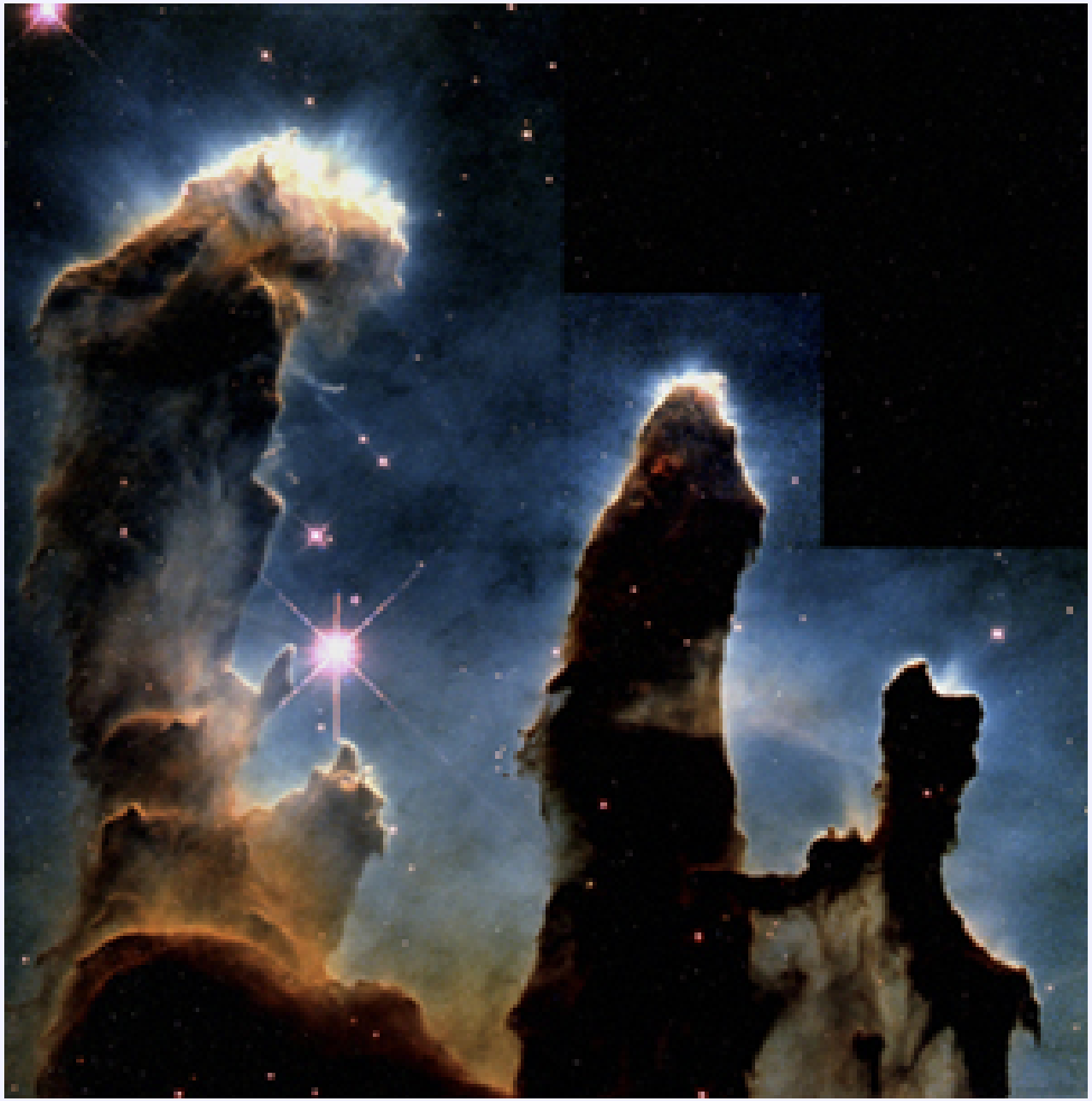
# Star formation

- Star formation happens as a result of a molecular cloud (cloud of gas and dust) collapsing after becoming gravitationally unstable
- It is still happening today, including in our Milky Way
- Observed best in the infrared or sub-millimetre and even in the radio
- Hubble, JWST, ALMA are the best facilities out there mapping star formation as it is happening

# Eagle nebula - Hubble



Credit: Hester and Scowen, NASA

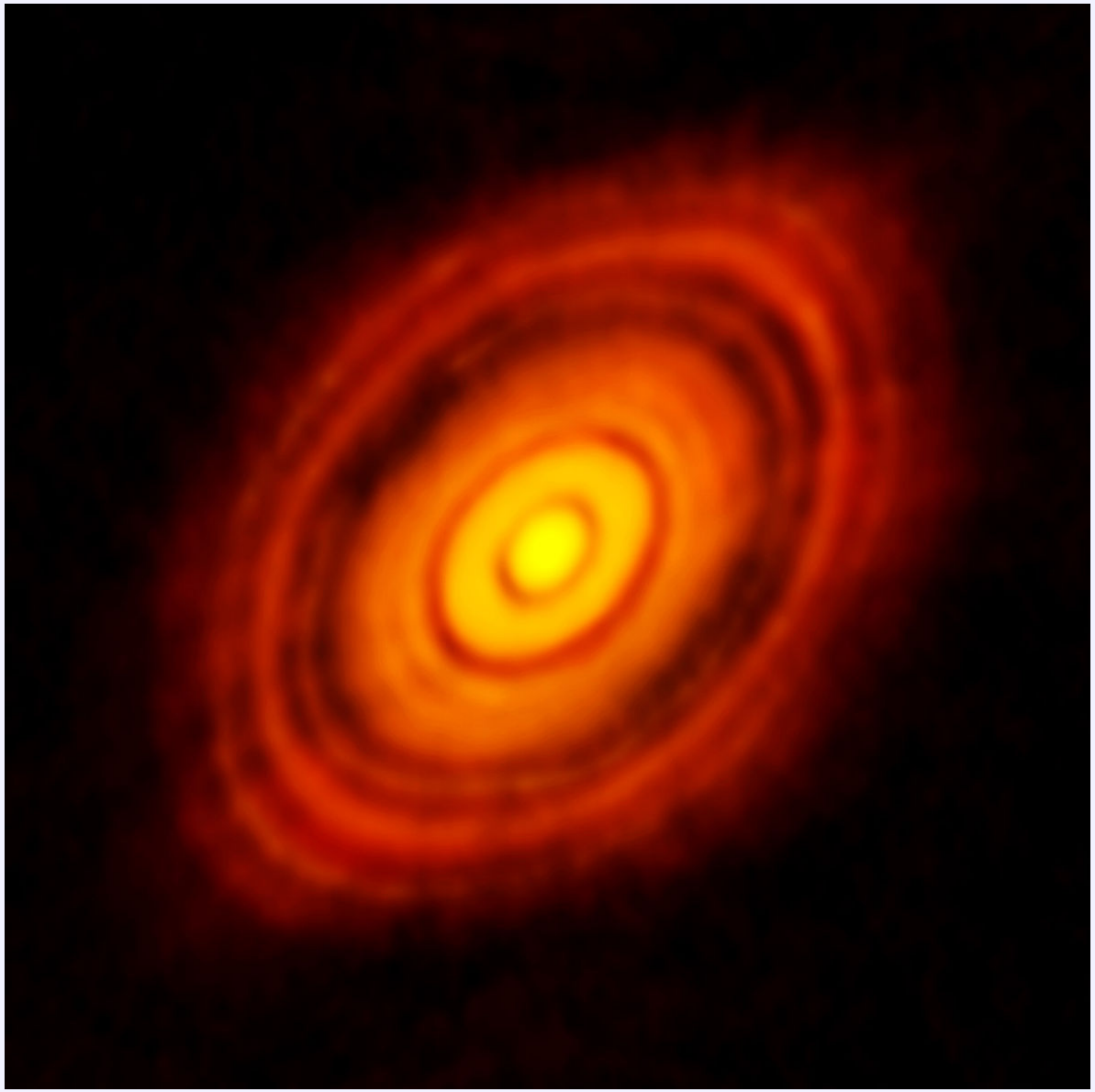
# Eagle nebula - Hubble vs JWST

Credit: NASA

# Protostar in Orion - JWST

# Protoplanetary disks

# Protoplanetary disks - the ALMA era



Credit: ALMA

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# Jeans instability

- Can we calculate the condition for gravitational instability that causes the cloud to collapse?
- Comes from a simple argument of energy balance (thermal and gravitational)
- Consider cloud of gas, mostly made up of hydrogen, with mass $M$, radius $R$
- The total energy of a hydrogen particle would be

$$E_{\rm tot} = \frac{3}{2}\, k\, T - \frac{G\, M\, m_u}{R}$$

- Cloud unstable towards collapse if $E_{\rm tot} < 0$

# Jeans length and Jeans mass

- We can rewrite the inequality to find that

$$R > \left( \frac{9\,k\,T}{8\pi\,G\rho\,m_u} \right)^{1/2} = R_J$$

  where $R_J$ is the Jeans length
- We can then find the Jeans mass, ie the mass corresponding to the Jeans length

$$M > M_J = \frac{4\pi}{3}\rho \left( \frac{9\,k\,T}{8\pi\,G\rho\,m_u} \right)^{3/2}$$

$$M_J = 2 \times 10^{-11}\,\mathrm{M}_\odot \left( \frac{T}{\mathrm{K}} \right)^{3/2} \left( \frac{\rho}{\mathrm{g\,cm^{-3}}} \right)^{-1/2}$$

- Typical gas clouds have $T = 100\,\mathrm{K}$ and $\rho = 10^{-22}\,\mathrm{g\,cm^{-3}}$, thus $M_J = 2000\,\mathrm{M}_\odot$, way too high.

# Fragmentation

- Initially, cloud is optically thin, meaning the temperature stays roughly constant
- As the cloud contracts, the cloud gets denser
- This isothermal contraction causes the Jeans mass to decrease
- Cloud thus becomes unstable on smaller and smaller scales $\rightarrow$ fragmentation (multiple stars form)
- Fragmentation also partly caused from rotational instability (see JCD 10.2)

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# Reaching hydrostatic equilibrium I

- Initially, cloud temperature roughly constant, hence, assuming an ideal gas
$$\left|\frac{dP}{dr}\right| \approx \frac{P}{R} \propto \frac{\rho}{R} \propto \frac{M}{R^4}$$
- At the same time, the term from gravitational force is
$$\frac{G\,M\,\rho}{R^2} \propto \frac{M^2}{R^5}$$
- Initially the gravitational force thus increases more rapidly than the pressure force

# Reaching hydrostatic equilibrium II

- When cloud density becomes high enough, contraction happens adiabatically and gas pressure can be described through electron degeneracy $P \propto \rho^{5/3}$ (remember lecture 4)
- The pressure force thus becomes proportional to

$$\left|\frac{dP}{dr}\right| \approx \frac{P}{R} \propto \frac{\rho^{5/3}}{R} \propto \frac{M^{5/3}}{R^6}$$

- At the same time, the term from gravitational force is still

$$\frac{G\,M\,\rho}{R^2} \propto \frac{M^2}{R^5}$$

- In this scenario, the pressure force thus increases more rapidly than the gravitational force
- This suggests that HE is indeed achievable

# Early star properties

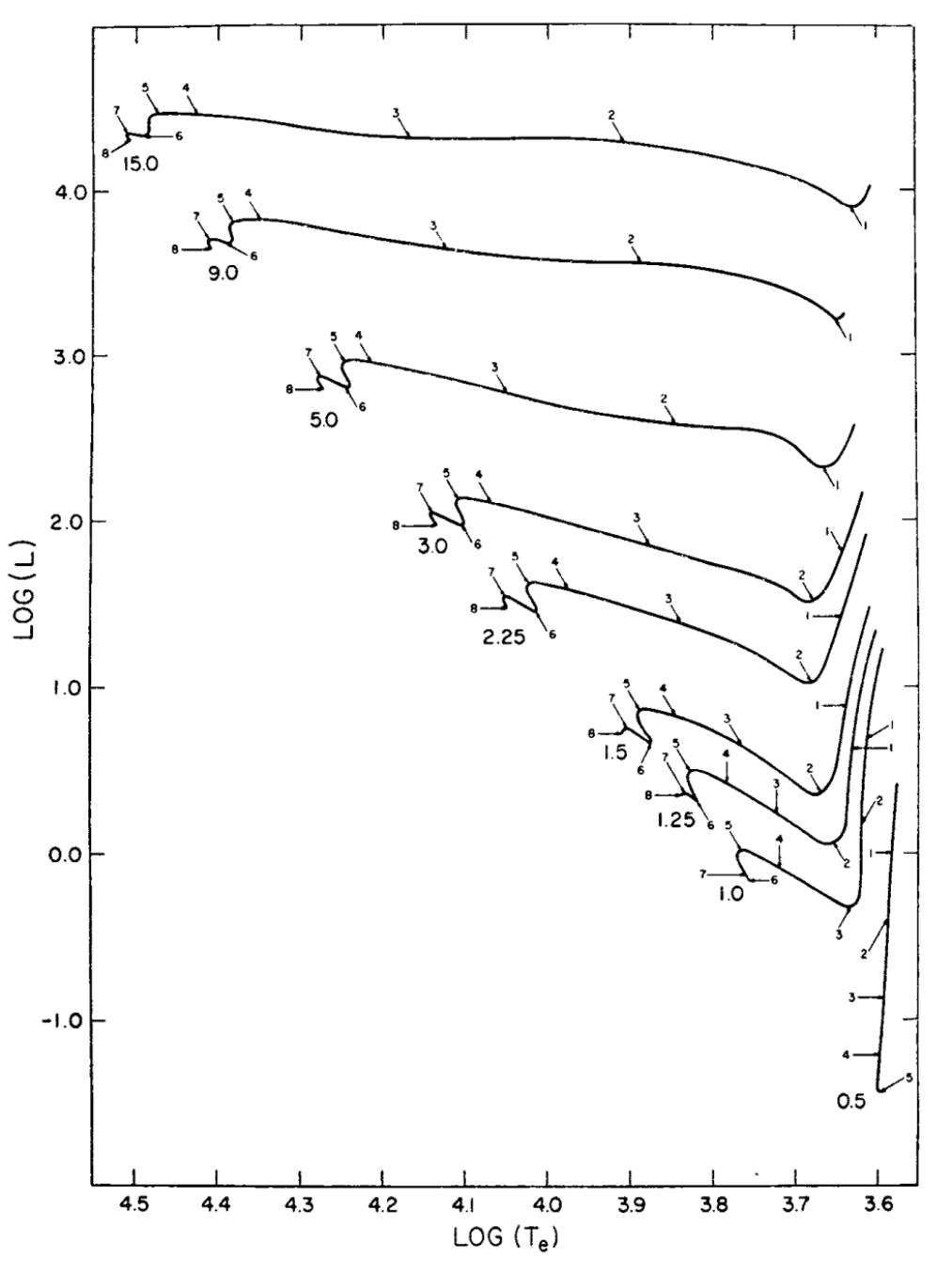
- When the star is in HE, the virial theorem (remember lecture 6) applies $E = \Omega/2$ where the luminosity of the star is caused by half the gravitational potential energy being released

$$L_G = -\frac{1}{2}\frac{d\Omega}{dt} = -\frac{1}{2}\frac{G\,M^2}{R^2}\frac{dR}{dt}$$

- Central temperature may be estimated (remember lecture 5) as

$$T_c = \frac{\mu_c m_u P_c}{k\rho_c} \approx \frac{G\,\mu_c m_u M}{k\,R}$$

- $R$ is still large so $T_c$ is still low hence the opacity (remember lecture 8) would be high
- It then follows that the condition is there for convective instability (lecture 10) meaning stars initially follow a Hayashi track




Credit: JCD, Fig 10.4

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# Minimum mass of a star

- Being a star means that hydrogen fusion must start
- For this core temperature must reach $\sim 10^7$ K
- With density increasing, a star can become degenerate (regions of high $\rho$, low $T$). Once that happens, increasing pressure no longer leads to an increase in temperature and fusion may never happen
- Detailed calculations show that $M_* > 0.08\,\mathrm{M}_\odot = 80\,\mathrm{M}_{\mathrm{Jup}}$.
- Protostars that evolve to bodies with lower mass, but still radiate [i.e. they are self-luminous], are called Brown Dwarfs

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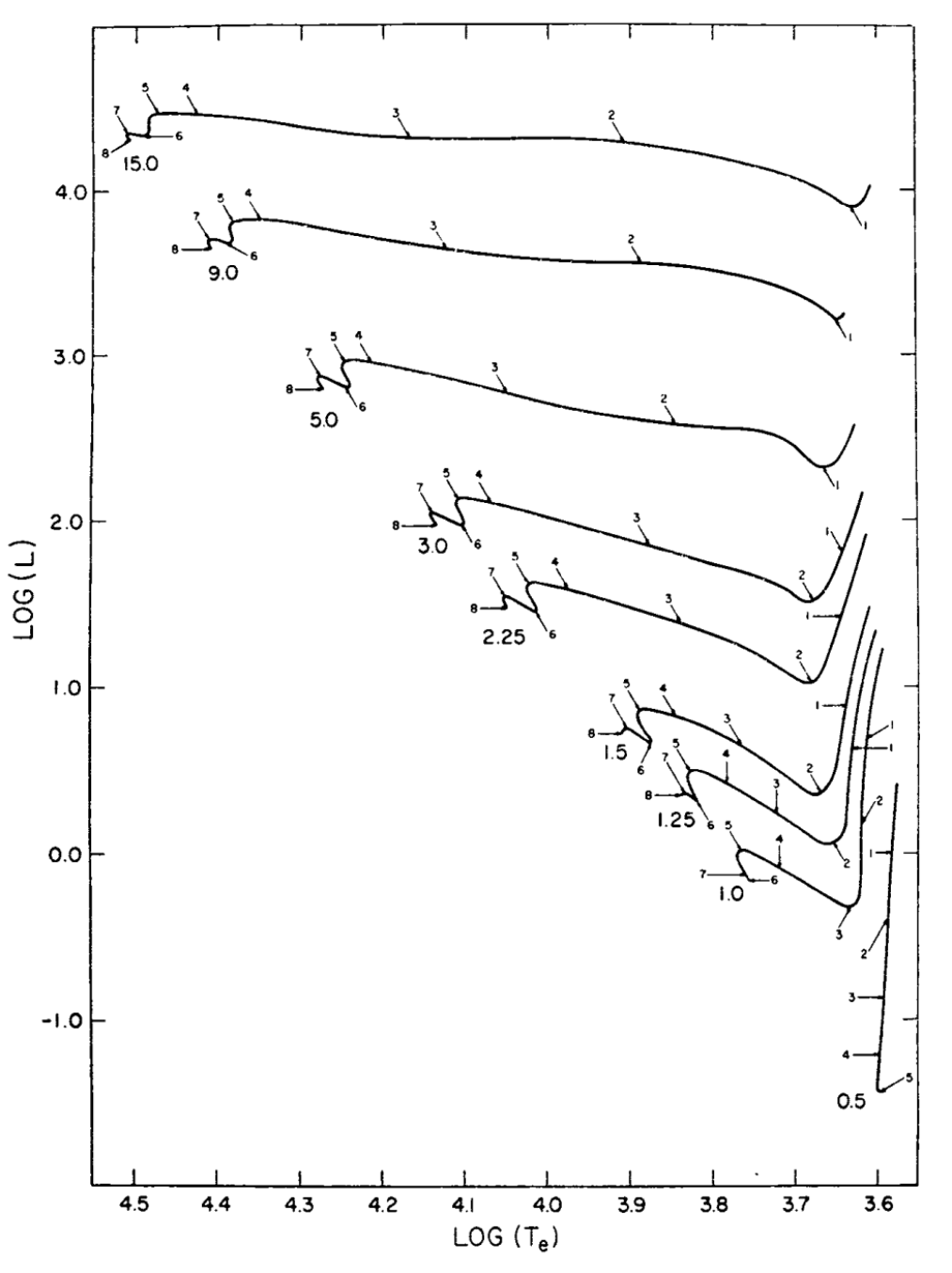


Credit: JCD, Fig 10.4

# Approach to the main sequence

- Stars with $M < 0.5\,\mathrm{M}_\odot$ continue to evolve via the Hayashi track
- More massive stars will eventually depart from the Hayashi track as their effective temperature will rise
- This will come with either a constant or increasing luminosity
- We call this the Henyey line

# Summary

- Star formation happens from gravitational contraction of molecuar gas clouds
- The Jeans mass is defined as the mass a cloud should at least have for collapse to happen

$$M_J = 2 \times 10^{-11}\,\mathrm{M}_\odot \left(\frac{T}{\mathrm{K}}\right)^{3/2} \left(\frac{\rho}{\mathrm{g\,cm^{-3}}}\right)^{-1/2}$$

- Stars can only start fusing hydrogen if they are more massive than $0.08\,\mathrm{M}_\odot$
- Stars initially evolve via a Hayashi track and then the Henyey line

NEXT LECTURE: The main sequence